

Error Log

Sr. No.	Line Number	Error Type	Error	Error Correction
1	10	Syntax Error	Missing semicolon after declaration	int department = 1;
2	12	Semantic Error	String literal assigned to int variable	int marks = 90;
3	26	Logical Error	Assignment operator '=' used instead of comparison operator	if(x == 10)
4	31	Scope/Semantic Error	Undeclared variable used	Declare score before use
5	50	Linker Error	Function called without declaration/definition	Declare and define add()
6	58	Scope Error	Variable used outside its block scope	Declare temp in outer scope or move printf inside block
7	62	Runtime Error	Array index out of bounds: arr[10]	Use index 0 to 4
8	66	Semantic Error	Wrong format specifier '%d' used for float	Use printf("%f\n", salary);
9	87	Scope Error	Variable used outside loop scope	Declare result before loop
10	91	Logical Error	Both if and else blocks produce same output	Modify one branch to intended output
11	98	Semantic Error	Required argument missing	displayData(value);
12	129	Logical Error	Extra semicolon causes infinite loop	Remove semicolon after while condition
13	134	Semantic Error	String literal assigned to char variable	char grade = 'A';
14	138	Semantic/Runtime Error	Wrong format specifier '%s' used for integer	Use printf("%d\n", number);
15	142	Runtime Error	NULL pointer dereferenced	Assign valid memory/address before dereferencing
16	146	Linker Error	Function called without declaration/definition	Declare and define calculateAverage()

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1	extern int total;	3	Linker Error	total is declared extern but never defined anywhere. Add a definition at file scope, e.g. int total = 0;
2	int marks = "90";	10	Type Mismatch Error	Cannot assign a string literal to an int. Use: int marks = 90;
3	printf("Program Started\n")	16	Syntax Error	Missing semicolon at the end of the statement.
4	int result = add(10,20);	36	Compile Error (Implicit Function Declaration)	add() is used before any prototype is declared. Add "int add(int a, int b);" before main(), or move the function definition above main().
5	printf("%d\n", &number);	42	Compile Error (Invalid Address-of Register Variable)	The address of a register-qualified variable cannot be taken. Remove "register", and if printing an address use %p with a cast: printf("%p\n", (void*)&number);
6	static int salary = "50000";	44	Type Mismatch Error	Cannot assign a string literal to an int. Use: static int salary = 50000;
7	printf("%d\n", department);	52	Scope Error	"department" was declared inside an inner block and is out of scope here. Declare it in the outer scope before use.
8	printf("%d\n", temp);	61	Scope Error	"temp" was declared inside the for-loop block and is out of scope here. Declare it before the loop.
9	arr[10] = 500;	65	Runtime Error (Array Index Out of Bounds)	arr has size 5 (valid indices 0-4). Use a valid index, e.g. arr[4] = 500;
10	printf("%d\n", a/b);	70	Runtime Error (Division by Zero)	b is 0, causing undefined behavior/crash. Check the divisor before dividing: if(b != 0) ...
11	if(value = 100)	74	Logical Error	Used assignment (=) instead of comparison (==): if(value == 100)
12	if(score > 80) ... else printf("Pass");	81	Logical Error	Both if and else branches print the same message. The else should print something different, e.g. "Fail".
13	char grade = "A";	96	Type Mismatch Error	A char must be assigned with single quotes, not a string literal. Use: char grade = 'A';
14	printf("%d\n", data);	118	Runtime Error (Uninitialized Variable)	data is used without being initialized, producing an indeterminate value. Initialize it first, e.g. int data = 0;
15	*ptr = 100;	122	Runtime Error (Null Pointer Dereference)	ptr is NULL; dereferencing it crashes the program. Allocate memory first, e.g. ptr = malloc(sizeof(int));
16	extern int companyCode;	124	Linker Error	companyCode is declared extern but never defined anywhere. Provide a definition, e.g. int companyCode = 0; at file scope.
17	average = calculateAverage(10,20);	130	Semantic Error (Undeclared/Undefined Function)	calculateAverage() is never declared or defined. Add a prototype and a matching function definition.
18	int employeeId = "EMP101";	134	Type Mismatch Error	Cannot assign a string literal to an int. Use a char array instead: char employeeId[] = "EMP101";
19	while(counter > 0);	138	Logical Error	Stray semicolon makes the while loop body empty, causing an infinite loop (counter is never decremented). Remove the semicolon: while(counter > 0) { counter--; }

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1	int choice	23	Syntax Error	Missing semicolon at the end of the declaration: int choice;
2	default	59	Syntax Error	A switch label needs a colon. Use: default:
3	if(index > 5)	85	Logical Error (Incorrect Bounds Check)	students[] has size 5 (valid indices 0-4), but the check allows index to reach 5 before flagging "Database Full", causing an out-of-bounds write. Check before writing and use: if(index >= 5) { printf("Database Full\n"); return; }
4	for(i=0;i<=totalStudents;i++)	95	Logical Error (Off-by-One)	Using <= accesses one element past the last valid student. Use: for(i=0;i<totalStudents;i++)
5	return total / 0;	119	Runtime Error (Division by Zero)	Should divide by the number of students, not the literal 0. Use: return totalStudents > 0 ? total / totalStudents : 0;
6	int highest = students[0].marks;	124	Type Mismatch Error	marks is a float; storing it in an int silently truncates the decimal part. Declare as: float highest = students[0].marks;
7	int lowest = students[0].marks;	126	Type Mismatch Error	marks is a float; storing it in an int silently truncates the decimal part. Declare as: float lowest = students[0].marks;
8	totalStudent	148	Semantic Error (Undeclared Identifier)	Typo: the actual variable is "totalStudents". Correct it to: totalStudents
9	printGrade();	150	Semantic Error (Undeclared/Undefined Function)	printGrade() is never declared or defined anywhere in the file. Add a prototype and a matching function definition, or remove the call.
10	calculateRank();	152	Semantic Error (Undeclared/Undefined Function)	calculateRank() is never declared or defined anywhere in the file. Add a prototype and a matching function definition, or remove the call.
11	s.rollNo = "101";	159	Type Mismatch Error	rollNo is an int; cannot assign a string literal. Use: s.rollNo = 101;
12	s.marks = "90";	163	Type Mismatch Error	marks is a float; cannot assign a string literal. Use: s.marks = 90;
13	students[roll].marks = 100;	176	Runtime Error (Unvalidated Array Index)	roll comes directly from user input and is never checked against array bounds (0 to totalStudents-1) before use. Validate first: if(roll >= 0 && roll < totalStudents) students[roll].marks = 100;
14	students[roll] = NULL;	185	Compile Error (Type Mismatch)	students[roll] is a struct, not a pointer; NULL cannot be assigned to it. Also roll is never bounds-checked. Reset the struct instead, e.g. memset(&students[roll], 0, sizeof(struct Student)); after validating roll.
15	if(name == students[0].name)	194	Logical Error	This compares array addresses, not the string contents. Use: if(strcmp(name, students[0].name) == 0)
16	avg = calculateAverage();	204	Type Mismatch Error	calculateAverage() returns a float; storing it in an int (avg) truncates the decimal part. Declare as: float avg;
17	fp = fopen("student.txt", "r");	218	Logical Error (Wrong File Mode)	The file is opened in read-only mode ("r") but then written to with fprintf. Open it for writing instead: fp = fopen("student.txt", "w"); and check fp != NULL before using it.
18	fp = fopen("missing.txt", "r");	229	Runtime Error (Missing NULL Check)	If the file does not exist, fopen returns NULL, and the following fscanf(fp,...) call will crash. Add a check: if(fp == NULL) { printf("File not found\n"); return; }

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1	<code>if(totalAccounts > 10)</code>	88	Logical Error (Incorrect Bounds Check)	<code>accounts[]</code> has size 10 (valid indices 0-9), but the check allows <code>totalAccounts</code> to reach 10 before flagging "Database Full," causing an out-of-bounds write. Check before writing and use: <code>if(totalAccounts >= 10) { printf("Database Full\n"); return; }</code>
2	<code>for(i=0;i<=totalAccounts;i++)</code>	98	Logical Error (Off-by-One)	Using <code><=</code> accesses one element past the last valid account. Use: <code>for(i=0;i<totalAccounts;i++)</code>
3	<code>accounts[acc].balance = accounts[acc].balance + amount;</code>	123	Runtime Error (Unvalidated Array Index)	<code>acc</code> comes directly from user input and is never checked against valid bounds before use. Validate first: <code>if(acc >= 0 && acc < totalAccounts)</code> <code>accounts[acc].balance += amount;</code>
4	<code>if(accounts[acc].balance > amount) { printf("Insufficient Balance\n"); } else { withdraw }</code>	139	Logical Error (Inverted Condition)	The condition is backwards: it prints "Insufficient Balance" when funds ARE sufficient, and withdraws when they are NOT. Swap the branches: <code>if(accounts[acc].balance < amount) printf("Insufficient Balance\n"); else { accounts[acc].balance -= amount; }</code>
5	<code>totalAccount</code>	192	Semantic Error (Undeclared Identifier)	Type: the actual variable is "totalAccounts". Correct it to: <code>totalAccounts</code>
6	<code>calculateInterest();</code>	194	Semantic Error (Undeclared/Undefined Function)	No function named <code>calculateInterest()</code> is declared or defined; the actual function in the file is named <code>interestCalculator()</code> . Call: <code>interestCalculator();</code> or rename the function.
7	<code>a.accNo = "1001";</code>	203	Type Mismatch Error	<code>accNo</code> is an int; cannot assign a string literal. Use: <code>a.accNo = 1001;</code>
8	<code>a.balance = "50000";</code>	207	Type Mismatch Error	<code>balance</code> is a float; cannot assign a string literal. Use: <code>a.balance = 50000;</code>
9	<code>accounts[accNo].balance = accounts[accNo].balance + 1000;</code>	218	Runtime Error (Unvalidated Array Index)	<code>accNo</code> comes directly from user input and is never checked against valid bounds before use. Validate first: <code>if(accNo >= 0 && accNo < totalAccounts)</code> <code>accounts[accNo].balance += 1000;</code>
10	<code>accounts[accNo] = NULL;</code>	228	Compile Error (Type Mismatch)	<code>accounts[accNo]</code> is a struct, not a pointer; <code>NULL</code> cannot be assigned to it. Also <code>accNo</code> is never bounds-checked. Reset the struct instead, e.g. <code>memset(&accounts[accNo], 0, sizeof(struct Account));</code> after validating <code>accNo</code> .
11	<code>if(name == accounts[0].name)</code>	237	Logical Error	This compares array addresses, not the string contents. Use: <code>if(strcmp(name, accounts[0].name) == 0)</code>
12	<code>average = generateAverage();</code>	247	Semantic Error (Undeclared/Undefined Function)	<code>generateAverage()</code> is never declared or defined anywhere in the file. Define it, or remove/replace the call.
13	<code>fp = fopen("accounts.txt","r");</code>	256	Logical Error (Wrong File Mode)	The file is opened in read-only mode ("r") but then written to with <code>fprintf</code> . Open it for writing instead: <code>fp = fopen("accounts.txt","w");</code> and check <code>fp != NULL</code> before using it.
14	<code>fp = fopen("missing.txt","r");</code>	267	Runtime Error (Missing NULL Check)	If the file does not exist, <code>fopen</code> returns <code>NULL</code> , and the following <code>fscanf(fp,...)</code> call will crash. Add a check: <code>if(fp == NULL) { printf("File not found\n"); return; }</code>
15	<code>arr[10] = 500;</code>	311	Runtime Error (Array Index Out of Bounds)	<code>arr</code> has size 5 (valid indices 0-4). Use a valid index, e.g. <code>arr[4] = 500;</code>
16	<code>*ptr = 100;</code>	315	Runtime Error (Null Pointer Dereference)	<code>ptr</code> is <code>NULL</code> ; dereferencing it crashes the program. Allocate memory first, e.g. <code>ptr = malloc(sizeof(int));</code>
17	<code>printf("%d\n",a/b);</code>	320	Runtime Error (Division by Zero)	<code>b</code> is 0, causing undefined behavior/crash. Check the divisor before dividing: <code>if(b != 0) printf("%d\n", a/b);</code>

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1	int choice	25	Syntax Error	Missing semicolon at the end of the declaration: int choice;
2	default	65	Syntax Error	A switch label needs a colon. Use: default:
3	if(totalBooks > 20)	90	Logical Error (Incorrect Bounds Check)	books[] has size 20 (valid indices 0-19), but the check allows totalBooks to reach 20 before flagging "Library Full," causing an out-of-bounds write. Check before writing and use: if(totalBooks >= 20) { printf("Library Full\n"); return; }
4	for(i=0;i<=totalBooks;i++)	100	Logical Error (Off-by-One)	Using <= accesses one element past the last valid book. Use: for(i=0;i<totalBooks;i++)
5	books[id].price = books[id].price - 10;	123	Runtime Error (Unvalidated Array Index)	id comes directly from user input and is never checked against valid bounds before use. Validate first: if(id >= 0 && id < totalBooks) books[id].price -= 10;
6	books[id].price = books[id].price + 10;	132	Runtime Error (Unvalidated Array Index)	id comes directly from user input and is never checked against valid bounds before use. Validate first: if(id >= 0 && id < totalBooks) books[id].price += 10;
7	totalBook	173	Semantic Error (Undeclared Identifier)	Typo: the actual variable is "totalBooks". Correct it to: totalBooks
8	calculateFine();	177	Semantic Error (Undeclared/Undefined Function)	calculateFine() is never declared or defined anywhere in the file. Add a prototype and a matching function definition, or remove the call.
9	b.bookId = "101";	184	Type Mismatch Error	bookId is an int; cannot assign a string literal. Use: b.bookId = 101;
10	b.price = "500";	190	Type Mismatch Error	price is a float; cannot assign a string literal. Use: b.price = 500;
11	if(title == books[0].title)	201	Logical Error	This compares array addresses, not the string contents. Use: if(strcmp(title, books[0].title) == 0)
12	books[id] = NULL;	213	Compile Error (Type Mismatch)	books[id] is a struct, not a pointer; NULL cannot be assigned to it. Also id is never bounds-checked. Reset the struct instead, e.g. memset(&books[id], 0, sizeof(struct Book)); after validating id.
13	books[id].price = books[id].price + 100;	222	Runtime Error (Unvalidated Array Index)	id comes directly from user input and is never checked against valid bounds before use. Validate first: if(id >= 0 && id < totalBooks) books[id].price += 100;
14	avg = calculateAveragePrice();	230	Semantic Error (Undeclared/Undefined Function)	calculateAveragePrice() is never declared or defined anywhere in the file. Define it, or remove/replace the call.
15	fp = fopen("library.txt", "r");	239	Logical Error (Wrong File Mode)	The file is opened in read-only mode ("r") but then written to with fprintf. Open it for writing instead: fp = fopen("library.txt", "w"); and check fp != NULL before using it.
16	fp = fopen("missing.txt", "r");	250	Runtime Error (Missing NULL Check)	If the file does not exist, fopen returns NULL, and the following fscanf(fp,...) call will crash. Add a check: if(fp == NULL) { printf("File not found\n"); return; }
17	*ptr = 100;	280	Runtime Error (Null Pointer Dereference)	ptr is NULL; dereferencing it crashes the program. Allocate memory first, e.g. ptr = malloc(sizeof(int));
18	arr[10] = 200;	284	Runtime Error (Array Index Out of Bounds)	arr has size 5 (valid indices 0-4). Use a valid index, e.g. arr[4] = 200;
19	printf("%d\n", a/b);	289	Runtime Error (Division by Zero)	b is 0, causing undefined behavior/crash. Check the divisor before dividing: if(b != 0) printf("%d\n", a/b);
20	strcpy(name, "Programming");	296	Runtime Error (Buffer Overflow)	name is only 5 bytes, but "Programming" needs 12 bytes (11 chars + null terminator). Enlarge the buffer: char name[20]; or use strncpy with a size limit.
21	ptr = malloc(5);	306	Runtime Error (Insufficient Memory Allocation)	Only 5 bytes are allocated, but the code writes 20 ints (80 bytes) into ptr, causing a heap buffer overflow. Allocate enough space: ptr = malloc(20 * sizeof(int));
22	printf("%d\n", ptr[0]);	315	Runtime Error (Use After Free)	ptr was already freed on the line above; accessing it afterward is undefined behavior. Read the value before calling free(ptr), or remove this access.
23	free(ptr);	317	Runtime Error (Double Free)	ptr was already freed earlier in this function; freeing it again is undefined behavior and can corrupt the heap. Remove this duplicate free(ptr) call.